

VPA IB DP CS — SL Revision + IA Development (Year 2 Semester 1)

SL Revision and IA Development SL复习与IA开发 SL Fùxí yǔ IA Kāifā

Target: Standard Level (SL) Students | Duration: 18 weeks parallel with HL topics

Programme Structure

While HL students study Topics 5 and 6, SL students follow a structured revision programme covering all four SL topics (1-4) while simultaneously developing their Internal Assessment. This ensures SL students are exam-ready while completing their IA to the highest standard.

Weeks 1 - 9: Topic 1 - 2 Revision + IA Planning

Week 1 — Diagnostic and Topic 1 Revision: Systems

Lesson	Focus	Activities
1	Diagnostic test: Topics 1-4 (IGCSE + IB baseline)	Identify weak areas; set personal targets
2	System components and stakeholders	Retrieval quiz; concept mapping
3	System life cycle and changeover methods	Scenario cards; justify changeover choice
4	ER diagrams and system modelling	Construct ER from description; peer mark
5	IA: Finalise client and problem statement (Criterion A)	One-to-one tutorial; sign-off

Week 2 — Topic 2 Revision: Computer Organization

Lesson	Focus	Activities
1	CPU architecture and FDE cycle	Trace FDE with given instructions
2	Memory hierarchy: RAM, ROM, cache, virtual memory	Comparison table; justify cache importance
3	Secondary storage and OS functions	Evaluate storage for given scenario
4	Binary representation: integers, two's complement, floating-point	Calculation practice; error analysis
5	Logic gates and Boolean algebra	Circuit design; truth table construction

Week 3 — Topic 1 - 2 Exam Practice

Lesson	Focus	Activities
1	Short-answer technique (1–4 marks)	Timed drills; peer marking
2	Explain and describe questions (5–8 marks)	Plan responses; write under timed conditions
3	Evaluate and discuss questions (10–12 marks)	Essay planning; balanced argument practice
4	Mock: Topic 1–2 paper (40 minutes, 30 marks)	Formal conditions
5	Mock feedback; DIRT	Markscheme analysis; set targets

Week 4 — Topic 3 Revision: Networks

Lesson	Focus	Activities
1	Network types, topologies, and hardware	Diagram practice; VPA network audit
2	Protocols and packet switching	Packet journey trace; protocol matching
3	Client-server, P2P, cloud, VPN	Comparison essay plan
4	Network security: encryption, firewalls, authentication	Security audit case study
5	IA: Design document workshop (Criterion B)	System diagram; algorithm design; UI mockups

Week 5 — Topic 4 Revision: Computational Thinking

Lesson	Focus	Activities
1	Pseudocode conventions and variables	Write pseudocode for standard problems
2	Selection and iteration structures	Trace table practice; predict outputs
3	Arrays, records, and collections	Data structure selection exercises
4	Searching and sorting algorithms	Efficiency comparison; trace practice
5	IA: Success criteria refinement	Make criteria measurable and testable

Week 6 — Topic 3–4 Exam Practice

Lesson	Focus	Activities
1	Programming exercises: Paper 2 style	Timed pseudocode writing
2	Trace table and algorithm analysis	Complete trace tables; identify errors
3	Mock: Topic 3–4 paper (50 minutes, 40 marks)	Formal conditions
4	Mock feedback; common errors analysis	DIRT; error log update
5	IA: Record of tasks initiation	Set up documentation template

Week 7 — Mixed Revision 1

Lesson	Focus	Activities
1	Topics 1–2 mixed questions	Retrieval practice; weak area focus
2	Topics 3–4 mixed questions	Retrieval practice; weak area focus
3	Cross-topic questions (e.g., network security + binary)	Synthesis practice
4	Exam technique: time management	Speed practice; question triage
5	IA: Coding workshop — core functionality	Pair programming; debug clinic

Week 8 — Mixed Revision 2

Lesson	Focus	Activities
1	Paper 1 full mock (Section A + B practice)	Timed section practice
2	Paper 2 full mock (programming problem)	Timed programming exercise
3	Self-assessment: mark own work with markscheme	Develop assessment literacy
4	Targeted revision: student choice	Independent study with teacher support
5	IA: Design document submission (Criterion B)	Peer review; teacher feedback

Week 9 — Consolidation and IA Development

Lesson	Focus	Activities
1	Revision clinic: student-requested topics	Q&A; problem solving
2	Revision clinic continued	Individual support
3	Assessment: SL Revision Test 2 (Topics 3–4)	Formal conditions
4	Test feedback; target update	DIRT; revise error log
5	IA: Development sprint — core features	Supervised coding; progress check

Weeks 10–18: Full Syllabus Revision + IA Completion

Weeks 10–12: Intensive IA Development

Lessons split between supervised IA development and targeted revision:

Monday/Tuesday: Supervised IA coding time with teacher circulation

Wednesday: IA troubleshooting clinic (common errors, debugging support)

Thursday: Topic rotation revision (Topics 1–4 cycle)

Friday: Retrieval quiz + IA progress check

Weeks 13–15: IA Testing and Evaluation + Exam Preparation

Lesson Focus	IA Activity	Revision Activity
Week 13	Testing strategy; test data generation; test table completion (Criterion D)	Paper 1 Section A: short answer speed drill
Week 14	User evaluation; client feedback; success criteria evaluation (Criterion E)	Paper 1 Section B: essay planning and writing
Week 15	Video evidence recording; final documentation check	Paper 2: programming under timed conditions

Weeks 16 – 18: Final Revision and Mock Examinations

Week	Focus
16	Full Paper 1 mock (1h30); full Paper 2 mock (1h30); IA final submission
17	Mock feedback; gap analysis; personalised revision plans
18	Semester 1 examination (Paper 1, SL 1h30); final IA submission deadline

SL Revision Resources

VPA Bilingual Glossary (180 terms)

Topic 1 – 4 condensed notes (one page per topic)

Past paper question bank sorted by topic and mark allocation

Trace table practice pack (20 problems)

Pseudocode writing prompts (30 problems)

Markscheme analysis guides

IA Support Schedule for SL Students

Week	Support Offered
1 – 2	One-to-one client confirmation tutorials
3 – 4	Design document workshop; peer review session
5 – 8	Weekly coding clinic (Tuesday after school)
9 – 12	Drop-in IA surgery (lunchtimes)
13 – 14	Testing and evaluation workshop
15	Video recording support; equipment booking
16 – 17	Final proofreading and formatting check

Differentiation for SL Revision

Level	Revision Focus	IA Support
Bronze	Secure 1–4 mark questions; accurate definitions; basic calculations	Structured templates for all IA sections; sentence starters
Silver	Secure 5–8 mark questions; clear explanations with examples	Guided design process; regular check-ins
Gold	Aim for 9–12 mark questions; evaluative depth; cross-topic synthesis	Independent development; peer mentoring; extension challenges

☐ Victoria Park Academy Shenzhen | IB DP Computer Science | SL Revision & IA Programme